

Lecture 17

Selection Sort

- It sorts by repeatedly selecting the smallest or largest element from the unsorted portion and swapping it with the first unsorted element.
- Algorithm
 1. Find the smallest element and swap it with first element.
 2. Find the second smallest from remaining elements and swap it with second element.
 3. Find the third smallest element from remaining elements and swap it with the third element
 4. Continue this process until the all elements moved to correct position.

Advantages & Disadvantages of Selection Sort

Advantages

- It requires constant extra space to sort the array.
- It requires less number of swaps compared to other sorting algorithms.

Disadvantages

- It has time complexity $O(n^2)$ makes it slower compared to Quick Sort or Merge Sort.
- Selection sort is not stable sorting algorithm.

Merge Sort

- Merge Sort follows divide & conquer approach.

Algorithm

1. **Divide:** keep dividing array into two halves until it cannot be divided.
2. **Conquer:** Each subarray is sorted individually using merge sort algorithm.
3. **Combine:** The sorted subarrays are merged back together in sorted order. This process continues until all the elements from both subarrays have been merged.

Time Complexity

- Best Case: when the array is already sorted or nearly sorted. $O(n \log n)$
- Average Case: when the array is randomly ordered. $O(n \log n)$
- Worst Case: when the array is sorted in reverse order. $O(n \log n)$

Application of Merge Sort

- Sorting large dataset
- External sorting, when dataset is too large to fit in memory.

Advantages of Merge Sort

- Merge Sort is a stable sorting algorithm.
- In worst case merge sort has time complexity $O(n \log n)$
- We can perform merging parallelly. Means this problem is suitable for parallel processing system.

Disadvantage of Merge Sort

- Merge sort require extra memory to store the merged sub-arrays during the sorting process.